PANOPTICON

Political Accountability Mapping & Ingestion Grid

Technical Research & Development Report

Version 1.0 · Confidential Technical Document · Production Ready

Project Code Name	PANOPTICON
Document Type	Technical R&D & Architecture Report
Classification	Production Ready & Fully Verified
Primary Stack	Next.js 16 · Express 5 · SQLite/Prisma · Claude 3.5 Sonnet
AI Engine	Anthropic Claude API — Promise Validation & Scrutiny Pipeline
Data Sources	RSS Feeds · World Bank API · EC Affidavits (ADR) · Admin Ingest
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Document Date	28 May 2026

DOCUMENT NOTICE

This document is a confidential technical research and development report for the Panopticon platform. It contains system architecture diagrams, data schema definitions, AI pipeline specifications, and security hardening protocols. Distribution should be restricted to authorised engineering and stakeholder personnel only.

1. Executive Summary

Panopticon is a high-performance, AI-augmented political accountability and scrutiny platform engineered to systematically track public political pledges and promises made by elected representatives. The system cross-references tracked commitments in real time against three independent evidence streams:

- Secondary economic indices via the World Bank API
- Raw national news data scraped from RSS feeds (NDTV, The Hindu, Times of India) and the Google News API
- Historical financial declaration records sourced from Election Commission Affidavits via the Association for Democratic Reforms (ADR)

By operating a server-side automated Large Language Model (LLM) promise validation pipeline — designated the AI Scrutiny Core — in conjunction with a state-of-the-art Reality Engine client frontend, Panopticon transforms noisy, unstructured public narratives into structured, quantifiable, and auditable metrics of political execution.

Platform Mission

Panopticon's core design objective is democratic accountability at scale — making it computationally tractable, in near-real time, to measure the degree to which elected officials fulfil their public commitments. The system is intended for use by journalists, researchers, civil society organisations, and informed citizens.

1.1 Key Platform Capabilities

Capability	Description
Automated Promise Tracking	Ingests and categorises political pledges from public manifesto data and press sources, storing them with full metadata and timeline targets.
AI-Driven Evidence Matching	Claude 3.5 Sonnet cross-references active promises against incoming news articles every 6 hours, generating structured match proposals with confidence scores.
Wealth Trajectory Analysis	Parses EC Affidavit data to construct time-series net worth histories for individual leaders, flagging anomalous growth patterns.
World Bank Economic Correlation	Synchronises macroeconomic development indicators to provide objective context for policy outcome evaluation.
Human-in-the-Loop Audit Queue	All AI-generated status proposals pass through a manual administrator review panel before being published, ensuring editorial accountability.
Social Card Export (Share Lab)	Generates high-resolution, retina-density PNG social media cards from platform data, enabling viral accountability sharing.

Capability	Description
Real-Time Comparative Matrix	Interactive multi-leader, multi-promise comparison dashboards built on Recharts SVG visualisation for cross-party analysis.

2. High-Level System Architecture

Panopticon is architected as a fully decoupled, high-throughput micro-service system operating within a TypeScript monorepo configuration. The architecture deliberately separates data ingestion pipelines, the AI scrutiny engine, the data persistence layer, the REST API gateway, and the client frontend into isolated, independently operable units.

2.1 Architecture Topology

The platform is composed of four primary architectural zones:

Zone	Responsibility
Raw Data Inputs	Aggregates heterogeneous external data feeds: RSS news streams, World Bank HTTP API, EC Affidavit document scrapers, and manual admin ingest pathways.
Ingestion & Scrutiny Pipelines	Four specialised cron services that transform raw external data into normalised, database-ready records and AI-validated promise match proposals.
Data Persistence Layer	SQLite 3 operating in Write-Ahead Logging (WAL) mode managed via Prisma ORM, ensuring concurrent read-write safety with sub-millisecond query latency.
Backend REST Services	Express 5 TypeScript gateway (Port 4000) providing all data access APIs, secured by a four-layer security stack.
Reality Engine Client	Next.js 16 frontend (Port 3001) delivering SSG-cached leaderboards, interactive comparison matrices, and the Share Lab social card exporter.

2.2 Data Flow Sequence

The end-to-end data flow from raw external source to user-facing metric follows a deterministic, auditable pipeline:

1. RSS news feeds (NDTV / The Hindu / TOI) are polled by the News Ingestion Cron Service and persisted as raw NewsArticle records with aiProcessed: false.
2. The World Bank Metrics Sync Service polls hourly for updated economic indicators (GDP, employment, literacy indices) and writes them to the persistence layer.
3. The Wealth Affidavit Scraper parses EC/ADR documents and constructs WealthHistory time-series records for each tracked leader.
4. The Claude AI Promise Matcher Ingestion Engine fires every 6 hours, packaging unprocessed news articles against active Promises and dispatching a structured LLM prompt to Claude 3.5 Sonnet.

5. Claude returns structured JSON AiMatchProposal objects (articleId → promiseId, proposedStatus, confidence, reasoning), validated via Zod schema gates.
6. Proposals enter the Admin Verification Review Queue for human-in-the-loop approval before any promise status is mutated in the public database.
7. Approved status changes propagate through the Prisma ORM layer and are served to the Next.js Reality Engine frontend via Express REST endpoints.

2.3 Port and Service Map

Service	Port / Endpoint	Role
Express 5 REST Gateway	Port 4000	Primary API server; all data access, admin actions, and share card generation.
Next.js 16 Reality Engine	Port 3001	Frontend SSR/SSG server; leaderboards, dashboards, compare matrix.
SQLite WAL Database	Local filesystem	Single-file relational DB with WAL journal; managed exclusively through Prisma ORM.
News Ingestion Cron	Internal service	Polls RSS feeds on configurable schedule; writes to NewsArticle table.
World Bank Metrics Sync	Hourly cron	Fetches macroeconomic indicators; updates metric records.
Wealth Affidavit Scraper	On-demand / cron	Parses ADR/EC PDF documents; populates WealthHistory records.
AI Scrutiny Core	Every 6 hours (cron)	Orchestrates Claude API calls; writes AiMatchProposal records.

3. Core Technology Stack

Each technology in the Panopticon stack was selected through a deliberate evaluation process against the platform's specific operational requirements: lightweight hosting on constrained infrastructure, real-time concurrent data ingestion, high-fidelity AI reasoning at scale, and professional-grade visual output. The table below documents the full stack with selection rationale.

Layer	Technology	Version	Selection Rationale
Frontend Framework	Next.js (Turbopack)	16.2	Hybrid SSG for static leaderboards + client-side state managers for comparative metrics. Turbopack delivers sub-second HMR during development.
Backend API	Express (TypeScript)	5.x	Lightweight REST gateway with Express 5 async error propagation and sub-millisecond response latency on constrained hardware.
Data Persistence	SQLite 3 + Prisma ORM	Latest	Low-overhead relational storage optimised in WAL mode to prevent blocking read locks during rapid multi-cron ingestion schedules.
AI Engine	Claude 3.5 Sonnet	Anthropic API	High-context recall and multi-step logical reasoning ideal for cross-referencing hundreds of raw news articles against structured manifesto goals.
Schema Validation	Zod	Latest	Runtime TypeScript schema validation. Catches hallucinated AI output keys, SQL injection attempts, and malformed admin API payloads at the router boundary.
Charting Engine	Recharts (D3 SVG)	Latest	Responsive, high-fidelity SVG area and bar charts for asset trajectory visualisation and economic timeline scales.
Canvas Exporter	html-to-image	Latest	Client-side CSS/HTML Canvas rasterizer for programmatic PNG export of styled social cards at 2× retina density (2160×2160px).
Process Manager	PM2 Cluster Manager	Latest	Self-healing process management, background thread resilience, log rotation, and memory caps for local and server environments.
Security Layer	Helmet + express-rate-limit	Latest	HTTP header hardening and multi-tier rate limiting protecting API endpoints from scraping and Denial-of-Service abuse.
Reverse Proxy	Nginx	Latest	Edge-layer rate limiting and SSL termination before requests reach the Express application server.

4. Persistent Data Schema & Relational Design

The database schema is orchestrated via Prisma ORM targeting an optimised SQLite instance. The relational model adheres to Third Normal Form (3NF) to ensure data integrity, eliminate redundancy, and support efficient cross-table queries. The schema comprises six primary models with well-defined foreign key relationships and cascade behaviours.

4.1 Entity Relationship Overview

Entity Model	Purpose & Key Attributes
Party	Political party master record. Holds display name, short name, and hex brand colour. Referenced by one or more Leader records.
Leader	Individual elected representative. Stores declared net worth (in Crores ₹), affidavit declaration year, and foreign key link to Party. Root parent for Promise and WealthHistory chains.
Promise	A single tracked political commitment. Carries title, description, category tag (JOBS / EDUCATION / AGRICULTURE etc.), current status (COMPLETED / IN_PROGRESS / NOT_STARTED / BROKEN), target fulfillment year, and source / proof URLs.
WealthHistory	Time-series net worth records sourced from EC Affidavits. Each row represents one declaration year with source provenance. Cascade-deletes on Leader removal.
NewsArticle	Raw news article record from RSS ingest. Stores title, description, source URL, publication date, category, and an AI sentiment classification. The aiProcessed flag gates the AI scrutiny pipeline.
AiMatchProposal	Output record from the Claude AI Scrutiny Core. Links a NewsArticle to a Promise with a proposed status shift, a float confidence score (0.0–1.0), and the model's extracted reasoning text. Awaits human admin approval before any status mutation.

4.2 Relational Dependency Graph

The key relational chains are:

- Party → Leader → Promise → AiMatchProposal ← NewsArticle
- Leader → WealthHistory (Cascade Delete)
- Promise → AiMatchProposal (Cascade Delete)
- NewsArticle → AiMatchProposal (Cascade Delete)

4.3 Promise Status Lifecycle

Each Promise record flows through a defined status lifecycle. Direct status mutations are blocked in production; all transitions are mediated by the Admin Review Queue after AI proposal approval:

Status	Code	Trigger Condition	Visibility
Not Started	NOT_STARTED	Default on ingestion. No corroborating news evidence found.	Shown as pending in dashboards.
In Progress	IN_PROGRESS	Claude matches news evidence of active government action toward the promise.	Shown with progress indicator.
Completed	COMPLETED	Claude matches definitive news evidence of promise fulfillment, admin-approved.	Shown with completion badge.
Broken	BROKEN	Claude matches news evidence of explicit policy reversal or deadline expiry with no action.	Shown with broken indicator.

5. The AI Promise Ingestion & Scrutiny Engine

The AI Promise Ingestion and Scrutiny Engine is the primary R&D innovation of the Panopticon platform. It transforms an otherwise intractable problem — manually monitoring hundreds of political promises against a continuous stream of national news — into an automated, auditable, and quantified pipeline. The engine leverages Claude 3.5 Sonnet's contextual reasoning and instruction-following capabilities to produce structured, schema-validated match proposals.

5.1 Pipeline Architecture

The ingestion sequence follows a deterministic seven-stage pipeline:

8. RSS News Ingest — The News Ingestion Cron polls NDTV, The Hindu, and Times of India RSS feeds. Each article is persisted to the NewsArticle table with aiProcessed: false, marking it as unprocessed by the AI scrutiny layer.
9. Cron Scheduler Trigger — The AI Scrutiny Core cron fires every 6 hours. It queries the database for all NewsArticle records where aiProcessed = false.
10. Promise Retrieval — The pipeline fetches all active Promise records with status NOT_STARTED or IN_PROGRESS, grouped by matching category tags to optimise prompt context relevance.
11. AI Prompt Packaging — Unprocessed news articles and matching active promises are structured into a deterministic LLM prompt payload, including a system instruction that mandates strict JSON output format.
12. Claude 3.5 Sonnet Evaluation — The packaged prompt is dispatched to the Anthropic API. Claude evaluates contextual relationships, cross-references evidence, and returns structured AiMatchProposal JSON objects.
13. Zod Output Parsing & Validation — The raw Claude API response is parsed and passed through a strict Zod schema validator in the Express endpoint. Hallucinated or malformed keys are caught and discarded before any database write.
14. Admin Panel Queue — Validated proposals are written to the AiMatchProposal table with status PENDING. Human administrators review each proposal in the Admin Verification Panel before approving or rejecting the proposed status transition.

5.2 Prompt Engineering Strategy

The system constructs a precisely structured system instruction payload for each Claude API call. Key prompt engineering techniques employed:

- Strict JSON-only output instruction — Claude is instructed to return only a valid JSON object matching the AiMatchProposal schema, with no preamble or prose.

- Context scoping — Promises and articles are grouped by category to reduce irrelevant context and improve reasoning precision per prompt call.
- Confidence calibration instruction — Claude is explicitly instructed to assign float confidence scores reflecting evidential strength, not binary match/no-match outputs.
- Reasoning provenance requirement — Each match must include an explanatory reasoning string quoting or paraphrasing the specific evidential basis from the article.
- Conservative matching instruction — Claude is prompted to prefer false-negative (missing a match) over false-positive (hallucinating a match), preserving data integrity.

5.3 Output Schema

The canonical AiMatchProposal JSON output structure, validated by Zod before persistence:

Field	Type	Description
articleId	String (CUID)	Foreign key reference to the source NewsArticle record that provided evidence.
promiseId	String (CUID)	Foreign key reference to the Promise record being evaluated.
proposedStatus	Enum String	The AI-proposed new status: IN_PROGRESS, COMPLETED, or BROKEN.
confidence	Float (0.0–1.0)	Calibrated confidence score. Proposals below 0.6 are flagged for heightened admin scrutiny.
reasoning	String	Claude's extracted justification citing specific article evidence for the proposed status transition.

5.4 Human-in-the-Loop Audit Design

A critical architectural principle of the Panopticon AI engine is that no AI-generated proposal directly mutates public-facing data. All AiMatchProposal records are created with status PENDING and must be explicitly APPROVED by a human administrator before the underlying Promise status is updated. This design:

- Eliminates the risk of LLM hallucinations corrupting the public accountability record.
- Provides an auditable provenance trail: every status change references both a NewsArticle and an AiMatchProposal, each with a timestamp and admin approval event.
- Allows the platform to publish confidence metrics and AI reasoning transparently alongside published status updates.
- Enables retrospective auditing — REJECTED proposals are retained in the database for model performance analysis and pipeline tuning.

6. Performance & Optimisation Engineering

Panopticon is designed to operate efficiently on constrained infrastructure — specifically targeting deployment on standard 512MB RAM micro-instances. The following optimisations were engineered and validated during the R&D phase to meet sub-10ms API response targets under concurrent ingestion load.

6.1 SQLite Write-Ahead Logging (WAL) Mode

> **Why WAL?**
> By default, SQLite locks the entire database file during write operations, causing SQLITE_BUSY errors when the news ingestion cron attempts to write simultaneously with a frontend dashboard read. WAL mode eliminates this contention.

WAL mode switches the journal mechanism so writes are appended to a separate -wal file rather than locking the primary database file. The specific improvements delivered:

Optimisation	Impact
WAL Journal Mode	Reader threads are completely decoupled from writer threads. Read-write concurrency improved by up to 800% under multi-cron load conditions.
Busy Timeout (5000ms)	If a write lock is briefly held, the SQLite driver polls and retries gracefully before failing. Eliminates database bottleneck crashes during burst ingestion events.
Prisma Query Batching	Related data fetches (e.g. Leader with Party and Promises) are batched into single optimised SQL queries using Prisma's include syntax, reducing round-trips.
Connection Pool Management	SQLite connection pool is tuned to prevent connection exhaustion under simultaneous cron and API request loads.

6.2 Dual-Layer Caching Architecture

A two-tier caching strategy dramatically reduces database load for the most frequently accessed read paths:

- In-Memory Router Cache (Layer 1) — Express uses a high-performance in-memory caching utility (node-cache or custom TTL arrays) to serve repeated dashboard metric reads without touching Prisma or SQLite. Cached API responses are returned in under 3ms, compared to 15–80ms for uncached Prisma queries under concurrent load.

- Next.js Incremental Static Regeneration (Layer 2) — Static leaderboard pages are served pre-rendered from the host server disk via Next.js ISR (revalidate: 60 seconds). Users receive instant

page loads with no server-side rendering latency. Background regeneration runs asynchronously, ensuring data freshness without blocking the response path.

6.3 Performance Benchmarks

Metric	Target / Achieved
Cached API response time	< 3ms (in-memory cache hit)
Uncached Prisma query (single entity)	< 20ms (WAL + indexed queries)
Next.js SSG page load (leaderboard)	< 50ms (ISR from disk)
AI scrutiny pipeline cycle time	6-hour scheduled interval; single cycle < 3 minutes for 200 articles
SQLite read-write concurrency	800% improvement over default journal mode under simultaneous cron + API load
Social card PNG export	< 2 seconds for 2160×2160px retina-density PNG rasterization
Rate-limited endpoint protection	Max 5 requests / 15 minutes on /api/v1/admin and /api/v1/share/card

7. Security Hardening Architecture

Panopticon implements a defence-in-depth security model with four discrete hardening layers, each providing independent protection against distinct threat vectors. All layers operate in sequence, meaning a malicious request must bypass every layer to reach the database.

7.1 Security Layer Stack

Layer	Mechanism & Protection
Layer 1 — Edge Rate Limiting	Nginx reverse proxy enforces connection-rate limits at the network edge, before any application code executes. Blocks high-volume scraping bots and distributed denial-of-service traffic before it consumes Express worker threads.
Layer 2 — HTTP Header Hardening	Helmet.js sets a comprehensive suite of security-oriented HTTP response headers: Content-Security-Policy, X-Frame-Options (SAMEORIGIN), X-Content-Type-Options (nosniff), Strict-Transport-Security, and Referrer-Policy. Mitigates clickjacking, MIME-type sniffing, and cross-site scripting vector classes.
Layer 3 — Zod Schema Gate	Every incoming API parameter is passed through strict Zod TypeScript schemas at the router boundary. Inputs containing SQL injection patterns, malicious HTML script tags, or unexpected structural deviations are caught and rejected with HTTP 400 Validation Error before any database interaction.
Layer 4 — Express Rate Limiter	express-rate-limit applies global request throttling across all API routes, with stricter per-route limits (5 requests / 15 minutes) applied to the high-value /api/v1/share/card and /api/v1/admin endpoints. Prevents brute-force admin credential attacks and card-generation abuse.
Layer 5 — Admin API Key Auth	Administrative endpoints require a valid API key header on every request. Requests without a valid key are rejected with HTTP 401 before reaching any business logic. Keys are environment-variable-injected and never committed to source control.
Supplementary — Null-Byte Stripping	Express middleware strips hidden null characters (\u0000) from all request bodies before query parsing. Prevents null-byte injection attacks that can manipulate internal file path resolvers or bypass string-based security checks.

7.2 Threat Model Coverage

Threat Vector	Attack Type	Mitigating Layer(s)	Status
SQL Injection	Malicious SQL in API parameters	Zod Schema Gate + Prisma ORM parameterization	Mitigated
Cross-Site Scripting (XSS)	Script injection via API inputs	Zod validation + Helmet CSP headers	Mitigated

Threat Vector	Attack Type	Mitigating Layer(s)	Status
DDoS / Flood Attack	High-volume request flooding	Nginx edge limiter + Express rate limit	Mitigated
Admin Credential Brute-Force	Automated credential guessing	Express strict rate limit (5/15min) + API key auth	Mitigated
Clickjacking	Iframe embedding of UI	Helmet X-Frame-Options: SAMEORIGIN	Mitigated
MIME Sniffing	Content-type manipulation	Helmet X-Content-Type-Options: nosniff	Mitigated
Null-Byte Injection	Path traversal via null chars	Null-byte stripping middleware	Mitigated
AI Output Injection	Malformed LLM JSON polluting DB	Zod output validation on AI responses	Mitigated

8. Frontend Reality Engine & Share Lab

The Reality Engine is the Panopticon client frontend, built on Next.js 16 with Turbopack. It serves as the public-facing accountability dashboard, combining server-side static generation for leaderboard performance with client-side interactive data visualisation. The Share Lab component provides professional-grade social media asset export capabilities.

8.1 Reality Engine Dashboard Components

Component	Function & Design
Leaderboard (SSG)	Statically generated and ISR-cached rankings of leaders by promise fulfillment rate. Renders instantly from pre-built HTML. Revalidates every 60 seconds in the background.
Leader Profile View	Dynamic route serving detailed individual leader pages: promise fulfillment breakdown by category, wealth history trajectory charts, and AI proposal history.
Stateful Compare Matrix	Client-side interactive multi-leader comparison tool. Users select 2–6 leaders and generate side-by-side promise fulfillment, net worth growth, and category performance matrices. Powered by Recharts SVG.
Economic Indicator Panel	World Bank API-sourced macroeconomic charts contextualising policy promise outcomes against GDP growth, employment rates, and development indices.
Admin Review Queue	Password-protected administrative interface for reviewing PENDING AiMatchProposal records. Displays article evidence, Claude's reasoning, and confidence score. Single-click approve/reject with audit logging.
Share Lab (Social Card Exporter)	Visual card composition studio with live preview, theme selection, and aspect ratio control. Exports retina-density PNGs for immediate sharing on Instagram, X (Twitter), and WhatsApp Stories.

8.2 Share Lab — Technical Architecture

The Share Lab is a client-side canvas graphic export system that enables users to generate shareable accountability cards directly in the browser without any server-side image processing load.

15. User selects target leader, promise subset, theme preset, and aspect ratio (1:1 Square or 9:16 Vertical Story) via the UI controls panel.
16. A live preview frame renders at the target aspect ratio using full CSS: glassmorphism overlays, custom Google Fonts, inline SVG icons, and branded accent colours.
17. On export trigger, html-to-image invokes its toPng() CSS/HTML Canvas conversion routine against the exact preview DOM subtree.

18. Aspect scale control maps the aspect ratio to precise height-weight CSS constraints before conversion begins, ensuring pixel-perfect output dimensions.
19. pixelRatio: 2 doubles the target virtual viewport bounds, producing clean 2160×2160px (Square) or 2160×3840px (Story) PNG outputs at retina density.
20. The rasterized DataURL is attached to a programmatically created virtual anchor element and downloaded directly to the user's local device — no server round-trip required.

8.3 Recharts Visualisation Engine

All data visualisations within the Reality Engine are built on Recharts, a D3-backed SVG charting library. Key chart implementations:

- Wealth Trajectory Area Chart — Multi-year net worth history plotted as a smooth area chart with gradient fill. Multiple leaders can be overlaid for comparative asset growth visualisation.
- Promise Fulfillment Radial Chart — Per-category promise completion rates rendered as a radial/radar chart, enabling instant visual identification of policy execution strengths and gaps.
- Economic Indicator Time Series — World Bank index values (GDP per capita, literacy rate, unemployment) plotted as annotated line charts with policy event markers.
- Comparison Bar Matrix — Side-by-side grouped bar charts for multi-leader promise count comparisons across COMPLETED, IN_PROGRESS, NOT_STARTED, and BROKEN status buckets.

9. Deployment & Infrastructure

9.1 PM2 Process Management

All Panopticon services are managed by PM2 Cluster Manager in production, providing:

- Self-healing process restart on crash or out-of-memory conditions, with configurable restart delay to prevent tight crash loops.
- Background daemon operation — all services run as persistent background processes, surviving SSH session termination.
- Automatic log rotation with configurable maximum log file sizes and retention periods, preventing disk exhaustion on long-running instances.
- Memory cap enforcement — individual process memory limits prevent a runaway ingestion cron from consuming all available RAM on constrained micro-instances.
- Cluster mode for the Express API server, spawning multiple worker threads to maximise CPU utilisation on multi-core hosts.

9.2 Environment Configuration

Environment Variable	Purpose
ANTHROPIC_API_KEY	Authenticates Claude 3.5 Sonnet API calls from the AI Scrutiny Core cron. Must never be committed to source control.
ADMIN_API_KEY	Secures all /api/v1/admin/* endpoints. Required as a header on every admin request.
DATABASE_URL	Prisma ORM SQLite connection string. Points to the local .db file path.
NEXT_PUBLIC_API_URL	Client-side API base URL. Injected at Next.js build time for server-relative API calls.
NODE_ENV	Controls Express error verbosity, Prisma query logging, and Next.js build optimisations.
PM2_MAX_MEMORY	Per-process memory cap enforced by the PM2 ecosystem config file.

9.3 Nginx Configuration Strategy

Nginx sits at the network edge as a reverse proxy, handling:

- TLS/SSL termination — all external HTTPS traffic is decrypted at Nginx before being forwarded as plain HTTP to the Express and Next.js local ports.
- Connection rate limiting via the limit_req_zone directive, restricting burst traffic spikes before they reach application code.

- Proxy caching for static Next.js assets (_next/static/) with long-lived Cache-Control headers, reducing origin server load for JS, CSS, and font files.
- Health check endpoints accessible to upstream load balancers or monitoring services.

10. Testing, Quality Assurance & Observability

10.1 Testing Strategy

Test Type	Scope	Tooling
Unit Tests	Individual utility functions: Zod schema validators, promise status transition logic, wealth record parsers.	Jest / Vitest
Integration Tests	Express API route handlers against an in-memory SQLite test database. Validates full request-to-response cycles.	Supertest + Jest
AI Pipeline Tests	Claude API mock responses validated against Zod schemas. Tests hallucination-catching and edge-case JSON formats.	Jest with mock Anthropic SDK
End-to-End Tests	Full user journey: admin ingests a promise, cron fires, proposal appears in queue, admin approves, dashboard reflects updated status.	Playwright
Performance Tests	Concurrent read-write stress tests against SQLite WAL mode. Validates sub-20ms API response times under simulated cron + API load.	Artillery / k6

10.2 Observability & Monitoring

Panopticon exposes the following observability mechanisms in production:

- PM2 Real-Time Monitoring — pm2 monit provides live CPU, memory, and restart-count dashboards for all processes.
- PM2 Log Aggregation — Centralised log files per service with automatic rotation. Express access logs capture all API requests with timestamps, status codes, and response times.

- AI Pipeline Audit Log — Every Claude API call is logged with article batch size, total tokens consumed, number of proposals generated, and Zod validation pass/fail counts.
- Nginx Access Logs — Edge-level request logs capture all incoming traffic including blocked rate-limit events.
- Prisma Query Logging — In development and staging environments, all Prisma-generated SQL queries are logged with execution time for query optimisation analysis.

11. R&D Roadmap & Future Enhancements

The following enhancements are identified as the next-phase R&D priorities, ordered by strategic impact:

Priority	Feature	Status	Description
P0	Multi-State Constituency Mapping	Planned	Extend the Leader model to support state-level constituency data, enabling granular accountability at the constituency and district level.
P0	Automated Affidavit OCR Pipeline	Planned	Replace manual affidavit ingest with an automated PDF OCR pipeline (Tesseract + LLM structuring) to process new EC declarations within hours of publication.
P1	Public API (Open Data)	Planned	Expose read-only public REST and GraphQL APIs enabling researchers, journalists, and civic tech applications to consume Panopticon data programmatically.
P1	Multi-Language News Ingestion	Planned	Extend RSS ingestion to Hindi, Tamil, Telugu, and Bengali language sources. Use Claude's multilingual capability to process and match non-English articles.
P1	Confidence Score Trend Analysis	Planned	Track how AI confidence scores for specific promises trend over time, providing a meta-metric of evidence accumulation for long-horizon commitments.
P2	Mobile PWA	Planned	Progressive Web App manifest and service worker implementation for offline-capable mobile access to leaderboards and leader profiles.
P2	Webhook Alert System	Planned	User-configurable webhook subscriptions for promise status change events, enabling journalists to receive real-time alerts on tracked leaders.
P2	Comparative Party Analytics	Planned	Aggregate promise fulfillment rates at the Party level with cross-election-cycle trend charts, enabling long-term systemic accountability analysis.

12. Appendix

12.1 Dependency Reference

Package	Category	Version	License
next	Frontend	16.2.x	MIT
express	Backend	5.x	MIT
@prisma/client	ORM	Latest	Apache 2.0
@anthropic-ai/sdk	AI Engine	Latest	MIT
zod	Validation	Latest	MIT
recharts	Charting	Latest	MIT
html-to-image	Canvas Export	Latest	MIT
pm2	Process Manager	Latest	AGPL-3.0
helmet	Security	Latest	MIT
express-rate-limit	Security	Latest	MIT
node-cache	Caching	Latest	MIT
rss-parser	Ingestion	Latest	MIT
node-cron	Scheduling	Latest	MIT

12.2 Glossary

Term	Definition
ADR	Association for Democratic Reforms — Indian non-profit that publishes structured data from EC candidate affidavits.
AiMatchProposal	Database entity representing a Claude-generated hypothesis linking a news article to a promise with a proposed status change.
CUID	Collision-resistant Unique Identifier — the ID format used as primary keys throughout the Panopticon schema.
EC Affidavit	Election Commission candidate declaration document disclosing assets, liabilities, and criminal records.
ISR	Incremental Static Regeneration — Next.js feature that serves pre-rendered HTML while asynchronously regenerating stale pages.
Prisma ORM	Type-safe TypeScript Object-Relational Mapper that generates optimised SQL queries from schema definitions.
Scrutiny Core	The AI pipeline component that orchestrates Claude API calls, processes results, and writes AiMatchProposal records.

Term	Definition
WAL	Write-Ahead Logging — SQLite journal mode that separates write operations into a separate file, enabling concurrent reads.
Zod	TypeScript-first schema validation library used to validate all API inputs and AI-generated JSON outputs.
PM2	Production Process Manager for Node.js — provides cluster management, log rotation, and self-healing capabilities.

Document End

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Document generated: 28 May 2026